Human Review: scalar $U(d, a)$ edge case

Original automatic proof: `solution_packet.pdf`
Checked date: 10 June 2026
Status: Manually checked.
Verdict: Rejected as a solution to the intended problem.

Human Comments

The calculation in `solution_packet.pdf` is a valid computation in the one-dimensional real Banach space $X = \mathbb{R}$. However, this is only a trivial scalar edge case. The obstruction comes from the fact that the unit sphere of $\mathbb{R}$ is just $\{-1, 1\}$, so for every $0 < d < 2$ the infimum defining $U(d, a)$ only sees the antipodal pair.

This is certainly not what was intended by the authors. The question was meant to probe genuinely small-distance geometry in Banach spaces with uniform property (S), not the degenerate behaviour of a two-point unit sphere. For that reason, the proposed solution should be rejected as an answer to the intended problem.

For the result to be relevant as a substantive contribution, one would need to address a nontrivial setting, for instance spaces whose unit sphere has admissible pairs at every small distance, or infinite-dimensional spaces with uniform property (S).

Short Version

This is a trivial one-dimensional obstruction and not the intended question. The packet should not be counted as resolving Open Problem (3).

Review Outcome

Reject this candidate as a solution to the intended problem. At most, archive it as a warning that automatic extraction can turn an intended geometric question into a degenerate scalar edge case.